Economic complexity – Homework 2

1. The Niche of Neffke, Otto & Weyh (2017)

The research problem is to understand how industry-specific human capital is and which industries have similar human capital requirements. Earlier relatedness measures, such as co-exports, co-occurrence, and input-output relations, capture similarities in products, locations, or value chains, but do not directly measure similarities in the skills required by industries. The authors ask whether inter-industry labor flows can reveal skill relatedness and whether this measure helps explain regional diversification and local industry growth. Using German social security data, they show that labor flows are concentrated in a small number of industry pairs, forming a sparse and stable network that is similar across worker types and independent of geographical distance. Their main contribution is demonstrating that skill relatedness predicts regional industry entry and growth better than co-location and value-chain relations, and providing relatedness matrices for future research.

2. Two citing articles (2020 or later)

(a) Sánchez-Moral (2022), Papers in Regional Science, "Skill-relatedness and regional economic development in Spain during the international crisis and the post-crisis period." Applies skill-relatedness to Spanish labor-market data, examining regional development during and after the economic crisis.

(b) Juhász, Elekes, Ilyés & Neffke (J. Econ. Geography, advance article; arXiv 2024), "Colocation of skill-related suppliers." Applies the same relatedness formula to Hungarian firm-to-firm labor flows and show how skill-relatedness reinforces the co-location of suppliers.

3. My niche on skill-relatedness or labor flows (related to previous homework on Instrumentational Complexity of Music Genres and Why Simplicity Sells)

Previous work of Neffke, Otto and Weyh (2017) showed that skill-relatedness between industries helps explain regional growth. Percino, Klimek and Thurner (2014) measured similarities between music genres using shared instruments, but did not examine artists' transitions between genres. It is therefore unclear whether shared instruments reliably capture skill-relatedness, which could help explain differences in genre growth.

Research question: Does skill-relatedness measured through artists' transitions between genres predict genre growth better than shared instruments do?

4. Data source and potential users

Data source: Discogs provides data on artists, genres, instruments and release dates. Artist transitions could measure relatedness, while changes in artist numbers could indicate growth, although album credits are an imperfect proxy for career moves.

Potential users: Record labels could identify potential transitions between genres, while arts funders and policymakers could support less-connected genres.